# Representations for Generation: A Survey of the Convergence of Visual Representation Learning and Generative Modeling

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## Abstract

Visual generation was long organized around a modular bargain: learn a perceptual compressor, freeze it, and train a generative model in its latent space. Representation learning developed in parallel, optimizing invariances for recognition rather than invertibility or likelihood. Since 2024 these lines have converged through representation-aligned denoisers, semantically shaped tokenizers, representation autoencoders, jointly learned latent priors, and native self-supervised generators. The resulting literature is difficult to compare because terms such as *representation*, *latent*, *end-to-end*, and *unified* conceal different gradient paths. We organize the field by two axes: where a representation enters the system, and how strongly representation and generation are coupled. We derive the objectives behind representative methods, isolate six properties of a useful generative latent—fidelity, semantics, information rate, modelability, geometry, and decoder robustness—and distinguish exact mathematical claims from empirical design hypotheses. We then connect the visual literature to language, multimodal systems, and molecular or 3D structure generation. Our main conclusion is deliberately non-monolithic: no single representation is universally best. The correct interface depends on which information should survive compression, which errors occur at inference, and which components are allowed to shape one another. This is a curated survey with a literature cutoff of 7 September 2026.

Preprint.

# 1 Introduction

Modern generative systems increasingly contain two learners that are easy to conflate. A *representation learner* maps data $x$ to features $z$ that expose useful factors of variation. A *generator* learns a probability distribution over $x$, or over a latent $z$ from which $x$ can be decoded. The former is usually rewarded for invariance; the latter ultimately pays for every visually consequential detail. Their objectives can cooperate, but they can also disagree.

Latent diffusion made this tension operational. A separately trained autoencoder first compresses images, and a diffusion or flow model then learns the latent distribution [Vahdat et al., 2021, Rombach et al., 2022]. This separation enabled efficient scaling, but it fixed the interface before the generator could express what it found easy or difficult. Meanwhile, self-supervised encoders such as DINO, DINOv2, MAE, and CLIP learned strong semantic features [Caron et al., 2021, Oquab et al., 2024, He et al., 2022, Radford et al., 2021]. The recent representations-for-generation literature asks what happens when these features supervise, initialize, replace, or co-evolve with the generative latent.

The literature is often narrated as a sequence: align denoiser features; align the autoencoder; discard the conventional VAE in favor of a foundation-model encoder; finally train representation and generation jointly. That chronology is useful, but not a taxonomy. REPA-E, UNITE, UNIFIED LATENTS (UL), and GENFIRST can all be called end-to-end while transmitting materially different gradients. SELF-FLOW learns internal features without an external vision foundation model (VFM), but does not thereby eliminate latent compression. LATENT FORCING co-denoises semantic features and pixels, yet does not learn a tokenizer. A model can share parameters without allowing every objective to update every representation.

This survey makes four contributions:

1. a two-axis taxonomy based on *intervention point* and *training coupling*;
2. a common mathematical language for auxiliary alignment, pretrained-feature latents, rate-controlled joint latent learning, and native self-supervision;
3. an evaluation framework that separates six latent properties and explicitly traces gradient flow; and
4. cross-domain case studies showing what changes in language, unified multimodal models, and molecular or 3D structure generation.

**Scope and evidence.** We focus on diffusion, flow-matching, and autoregressive systems for which learned representations are a primary design variable. This is a curated technical survey, not a systematic review or a leaderboard. Many 2026 results are preprints with incomparable compute, guidance, sampling, and evaluation protocols. We label a company announcement as product evidence rather than as a reproducible method, and use social-media posts only for discovery and context. Unless otherwise stated, numerical claims belong to the cited paper's own protocol.

**Central thesis.** The choice of latent is a constrained interface-design problem. A useful code must preserve what the decoder cannot cheaply hallucinate while presenting a distribution that the prior can learn. Reconstruction and recognition probe only parts of that contract. Information rate, covariance and spectral structure, manifold geometry, exposure to off-manifold inputs, and the exact gradient path can dominate.

# 2 A Common Mathematical Language

## 2.1 The modular latent generator

Let data $x \sim p_{\mathrm{data}}$, encoder $q_\phi(z \mid x)$, decoder $p_\psi(x \mid z)$, and learned prior $p_\theta(z)$. A modular latent generator performs

$$x \xrightarrow{q_\phi} z \qquad \text{and} \qquad z_0 \sim p_\theta \xrightarrow{p_\psi} \hat{x}.$$

The interface $z$ serves two clients with different wishes. The decoder prefers a code rich enough for faithful reconstruction. The prior prefers a regular, low-complexity distribution. A semantic learner may prefer invariance to texture, pose, or instance detail. Latent quality therefore cannot be a single scalar.

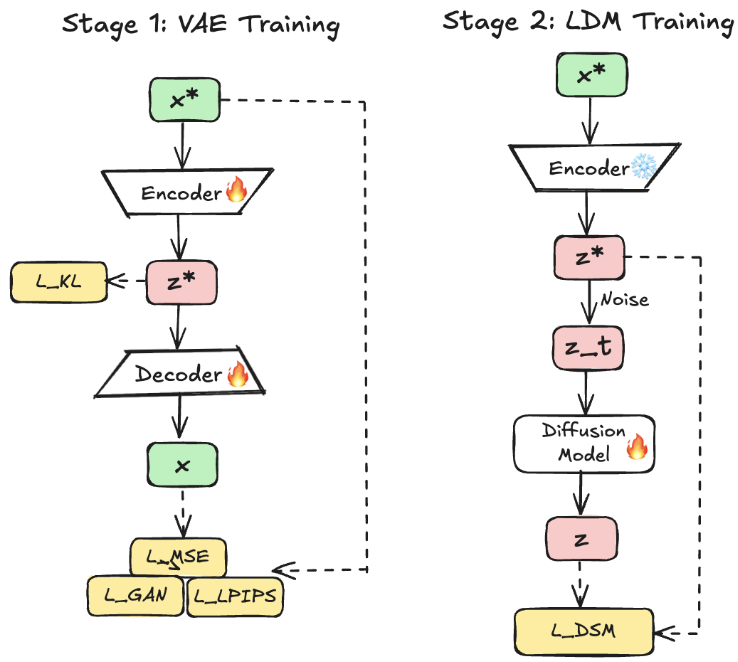



Figure 1: The modular latent-diffusion recipe: train the autoencoder and latent prior in separate stages. Author-created Excalidraw diagram based on the latent diffusion architecture [Rombach et al., 2022].

For a variational autoencoder (VAE), introduce an auxiliary posterior $q_\phi(z \mid x)$. The marginal log likelihood obeys

$$\begin{aligned}\log p_{\theta,\psi}(x) &= \log \int q_\phi(z \mid x) \frac{p_\theta(z) p_\psi(x \mid z)}{q_\phi(z \mid x)}\, dz \\ &\geq \underbrace{\mathbb{E}_{q_\phi(z|x)}[\log p_\psi(x \mid z)]}_{\text{reconstruction}} - \underbrace{D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z))}_{\text{rate / prior match}} \equiv \mathcal{L}_{\mathrm{ELBO}}(x),\end{aligned} \tag{1}$$

where the inequality is Jensen's. The color code will be used throughout: blue denotes encoder or representation terms, orange denotes prior or generative terms, green denotes decoder terms, and purple denotes external or target representations.

Equation (1) exposes the classic trade-off but hides two practical facts. First, perceptual and adversarial reconstruction losses need not correspond to a normalized likelihood. Second, a simple analytic prior such as $\mathcal{N}(0, I)$ can be replaced by a learned diffusion prior. The resulting system can be trained jointly, in stages, or with selected stop-gradients. These choices change the optimization problem even when the forward graph looks identical.

### 2.2 Diffusion and flow objectives

In a variance-preserving diffusion parameterization [Ho et al., 2020, Song et al., 2021],

$$z_t = \alpha_t z_0 + \sigma_t \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I), \qquad \alpha_t^2 + \sigma_t^2 = 1. \tag{2}$$

A denoiser may predict noise, clean data, score, or velocity. For noise prediction,

$$\mathcal{L}_{\mathrm{diff}}(\theta) = \mathbb{E}_{z_0,t,\epsilon}\left[w(t)\, \|\epsilon - \epsilon_\theta(z_t, t, c)\|_2^2\right], \tag{3}$$

with optional condition $c$. Different $w(t)$ correspond to different variational or perceptual emphases. The apparent simplicity of Equation (3) is deceptive: rescaling $z_0$, changing its covariance, or concentrating it near a manifold changes the effective signal-to-noise ratio and the geometry traversed by the probability path.

Flow matching uses a conditional interpolation, commonly [Lipman et al., 2023]

$$z_t = (1-t)z_{\text{noise}} + t z_{\text{data}}, \qquad u_t = z_{\text{data}} - z_{\text{noise}},$$

and minimizes

$$\mathcal{L}_{\text{FM}}(\theta) = \mathbb{E}\left[\|v_\theta(z_t, t, c) - u_t\|_2^2\right]. \tag{4}$$

The learned vector field defines the ODE $dz_t/dt = v_\theta(z_t, t, c)$. Straight Euclidean paths are convenient, not neutral: if codes occupy a curved or anisotropic set, interpolants may leave high-density regions.

### 2.3 Representation alignment

Let $h_\theta^\ell(z_t, t)$ be a hidden denoiser feature and $r_\omega(x)$ a frozen teacher feature. A representative alignment objective is

$$\mathcal{L}_{\text{align}} = \mathbb{E}\left[1 - \frac{g(h_\theta^\ell)^\top \operatorname{sg}(r_\omega(x))}{\|g(h_\theta^\ell)\|_2 \, \|\operatorname{sg}(r_\omega(x))\|_2}\right], \qquad \mathcal{L} = \mathcal{L}_{\text{diff}} + \lambda(t,k)\mathcal{L}_{\text{align}}. \tag{5}$$

The projector $g$, layer $\ell$, teacher, spatial correspondence, and schedule $\lambda$ are all substantive design choices. Because the teacher is stopped, alignment changes the denoiser, not the target representation. If $\lambda$ is switched off early, as in HASTE, the term is better understood as an optimization curriculum than as a permanent semantic constraint [Yu et al., 2024, Wang et al., 2025].

### 2.4 Trace the gradients, not the slogans

For a joint objective

$$\mathcal{J}(\phi, \theta, \psi) = \lambda_{\text{rec}}\mathcal{L}_{\text{rec}} + \lambda_{\text{prior}}\mathcal{L}_{\text{prior}} + \lambda_{\text{rep}}\mathcal{L}_{\text{rep}}, \tag{6}$$

the crucial question is whether

$$\nabla_\phi \mathcal{L}_{\text{prior}} \quad \text{is active, stopped, approximated, delayed, or mediated only by shared weights.}$$

If it is active, the generator can reshape the encoder. That can improve modelability, but it can also collapse the code to make prior fitting easy. If it is stopped, representation and generation may still interact through a shared trunk or through alternating tasks. Joint training should therefore be reported as a gradient-routing matrix, not a Boolean flag.

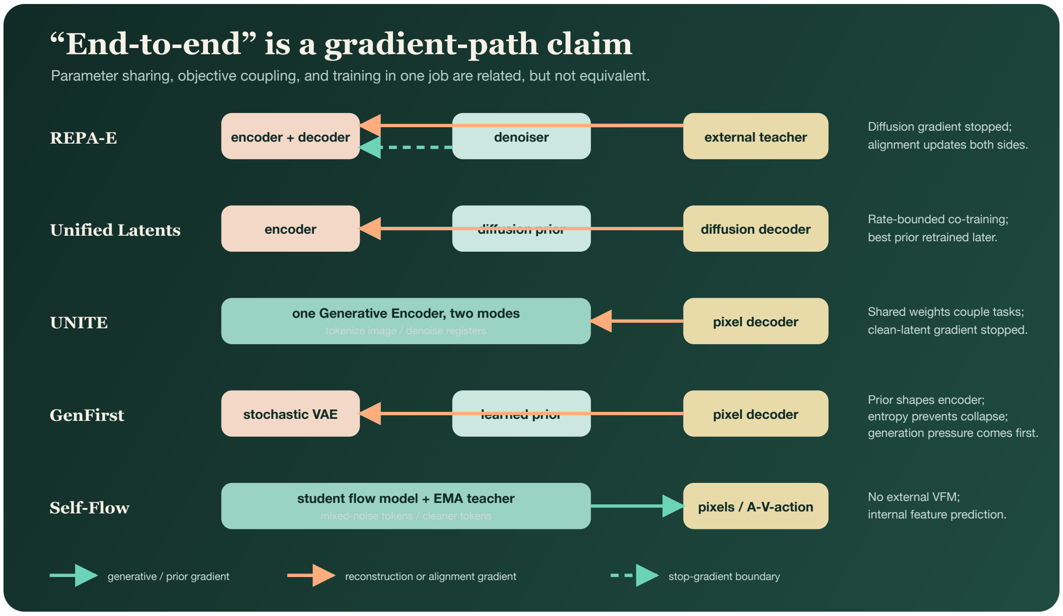



Figure 2: Original comparison of dominant gradient paths. Similar forward pipelines can implement different optimization problems. A stop-gradient on the clean code does not imply complete independence when parameters are shared elsewhere.

# 3 From Parallel Literatures to a Two-Axis Taxonomy

## 3.1 Historical compression

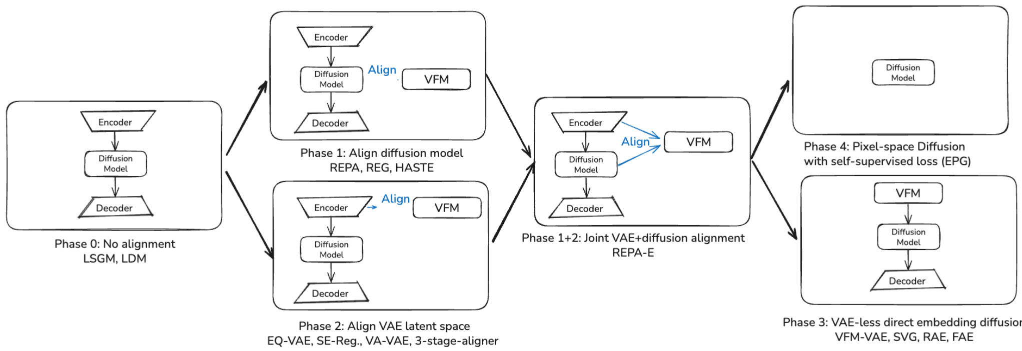
The VAE formalized amortized latent-variable learning [Kingma and Welling, 2014]; VQ-VAE supplied a learned discrete tokenizer [van den Oord et al., 2017]. Score-based latent models showed that a learned diffusion prior could be integrated with a hierarchical autoencoder [Vahdat et al., 2021]. Latent diffusion then established the modular engineering recipe: train a perceptual compressor, freeze it, and learn a denoiser in the compressed space [Rombach et al., 2022]. DiT made the latent denoiser a scalable Transformer [Peebles and Xie, 2023].

In parallel, self-supervised visual encoders developed increasingly transferable semantic spaces. Their features were usually evaluated by linear probes, retrieval, segmentation, or robustness, not by whether pixels could be reconstructed or whether a generative prior could model their full joint distribution. An important reverse-direction precedent was DDAE: rather than importing a representation teacher into a generator, it showed that intermediate features learned by unconditional diffusion pretraining already support linear probing and transfer [Xiang et al., 2023]. Contemporaneous work likewise studied diffusion models as masked autoencoders or representation learners [Wei et al., 2023, Yang and Wang, 2023]. These results did not yet solve the latent-interface problem, but they established that generation and representation learning need not be separate networks.

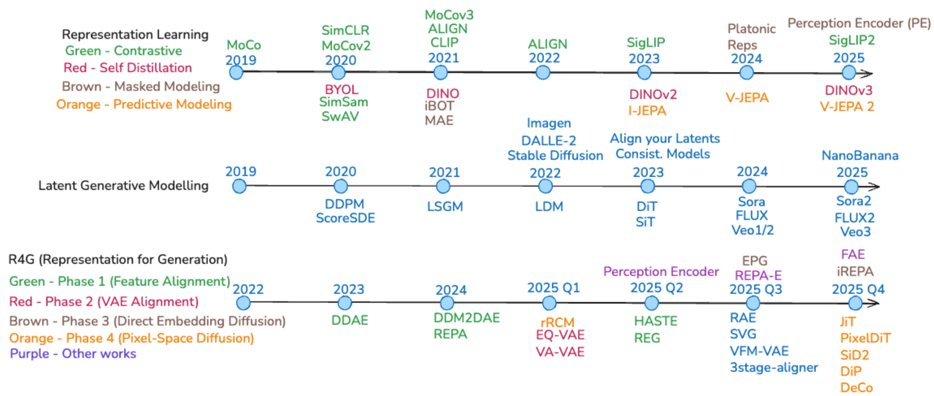
The convergence accelerated when generative models used external self-supervised features as internal targets. REPA aligned DiT hidden states to frozen semantic features and reported substantially faster optimization [Yu et al., 2024]. Follow-up work varied targets, schedules, and coupling, while another branch moved the alignment into the tokenizer. A third branch replaced the conventional VAE encoder with a VFM. The 2026 wave made the final step explicit: allow representation, prior, and decoder to shape one another.

## 3.2 The taxonomy

Figure 4 separates two questions.




(a) The original post's four-phase storyline: from unaligned latent generation to joint/native representation learning.




(b) A chronological snapshot of representation learning, latent generation, and representations-for-generation through the first RAE wave.

Figure 3: Two complementary views retained from the accompanying article. The phase view is a teaching narrative; the timeline records precedent. Neither replaces the mechanism-first taxonomy in Figure 4. Author-created Excalidraw diagrams.

**Where does the representation enter?** (i) as an auxiliary target inside a denoiser; (ii) as a constraint on a tokenizer or VAE; (iii) as the generative latent itself; or (iv) inside a native pixel/data-space or multimodal model. These positions imply different inference costs and different failure modes.

**How strongly is training coupled?** A representation can be frozen, selectively adapted or staged, or jointly learned. Parameter sharing, gradient sharing, and schedule coupling are distinct. A frozen teacher can still strongly shape early optimization; a shared encoder can receive stopped clean-code gradients; a joint first stage can still be followed by a freshly trained prior.

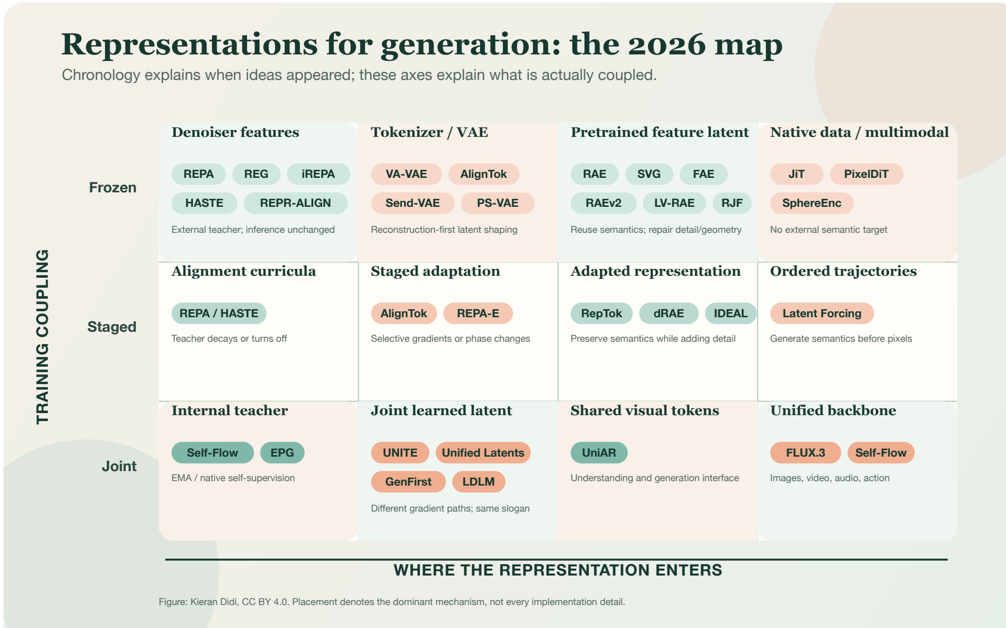



Figure 4: A two-axis map of representative methods. Placement denotes the dominant mechanism rather than every implementation detail. Chronology explains when ideas appeared; these axes explain what is coupled. Original figure, CC BY 4.0.

| Family | Representation enters as | Typical coupling | Inference consequence | Diagnostic question |
|---|---|---|---|---|
| Denoiser alignment | auxiliary hidden-state target | frozen teacher; scheduled weight | usually none | Does alignment accelerate useful features or overconstrain them? |
| Aligned tokenizer | regularizer or target for encoder | frozen teacher; encoder adapted | same latent pipeline | Which semantic invariances can reconstruction tolerate? |
| Representation AE | pretrained feature is the code | frozen/lightly adapted encoder | decoder must invert VFM space | Are semantic codes complete, normalized, and locally decodable? |
| Joint latent learning | learned interface among encoder, prior, decoder | shared or routed gradients | architecture dependent | Can prior pressure improve modelability without collapse? |
| Native/self-supervised | target generated by model or trajectory | joint/EMA | may remove external teacher | What asymmetry prevents a trivial target? |

Table 1: Mechanism-first comparison. The final column is more portable than any method label.

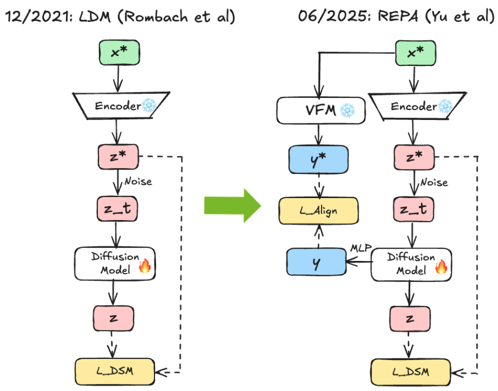
## 4 Auxiliary Representations Inside the Generator

REPA adds a semantic prediction task to intermediate denoiser features while leaving the VAE and inference graph unchanged [Yu et al., 2024]. This is attractive because the auxiliary loss is removable: one can improve optimization without paying for the teacher at sampling time. It also creates a clean experimental question about what feature target helps.

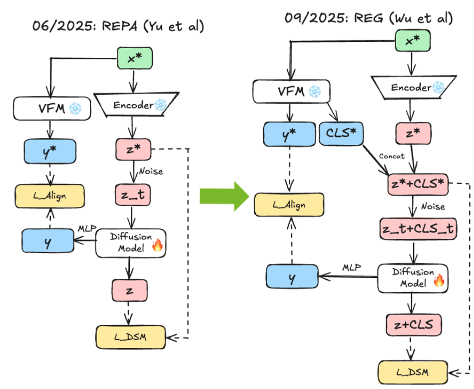
The answer is not simply the strongest recognition model. WHAT MATTERS FOR REPRESENTATION ALIGNMENT finds that spatial self-similarity is highly predictive in its controlled comparison, while global semantic quality alone is insufficient [Singh et al., 2025]. The result is specific to using a representation as a patchwise denoiser target. It does not imply that global semantics are unimportant when the representation is the latent itself. Indeed, RAEv2 reports benefits from globally semantic encoders while an added REPA-style objective improves spatial organization [Singh et al., 2026]. The location of the representation changes the desired property.

Permanent alignment can also become a liability. HASTE reports that turning alignment off early preserves its optimization benefit while avoiding late-stage conflict [Wang et al., 2025]. REG entangles representation guidance more deeply with generative features [Wu et al., 2025]. These variants suggest three interpretations of Equation (5): representation alignment can be (i) a semantic regularizer, (ii) a preconditioner or curriculum that shapes early features, or (iii) an internal guidance signal. Experiments should distinguish them by sweeping the active interval, target layer, spatial granularity, and teacher.

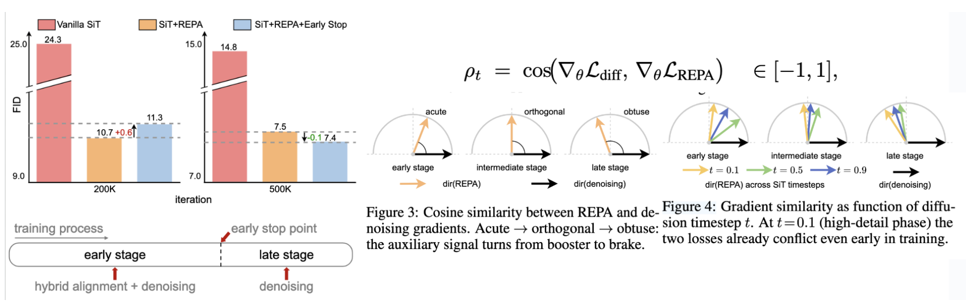
**Practical recipe.** Normalize target and predicted features, isolate the alignment head from the output head, and report whether the teacher operates on clean images, decoded latents, or augmented views. Log the cosine loss by diffusion time. If alignment helps only at high noise or only early in training, a scheduled objective is more faithful than an always-on penalty. Evaluate generation after removing the head to verify that the claimed gain is carried by the denoiser rather than by an inference-time crutch.




(a) REPA adds a removable VFM-alignment target to the denoiser.




(b) REG feeds representation-derived information more deeply into generation.




(c) HASTE turns alignment off after an early phase, separating feature acquisition from late denoising refinement.

Figure 5: The Phase-1 alignment lineage: REPA [Yu et al., 2024], REG [Wu et al., 2025], and HASTE [Wang et al., 2025]. The forward graph changes less than the duration and role of the representation signal. Author-created Excalidraw diagrams; the HASTE panel includes small adapted result panels from the cited work, which are excluded from this paper's CC BY license.

# 5 Shaping the Tokenizer and Adopting Pretrained Features

## 5.1 Semantically aligned autoencoders

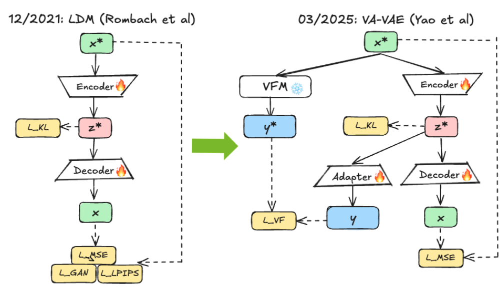
Conventional perceptual autoencoders already borrow representation losses through LPIPS-like feature distances, but recent methods place semantic organization at the center. REPA-E allows a representation objective to update the VAE encoder while stopping the diffusion loss from flowing into that encoder [Leng et al., 2025]. Thus it is end-to-end in scheduling and system assembly, not in the strongest gradient-sharing sense. ALIGNTOK adapts a VFM toward a compact tokenizer while trying to preserve useful semantic structure [Chen et al., 2025]. PS-VAE argues that both semantics and reconstruction are needed for text-to-image generation and editing [Zhang et al., 2025]. Work on autoencoder diffusability likewise shows that reconstruction FID can improve while the latent distribution becomes harder to model [Skorokhodov et al., 2025].

VA-VAE makes the reconstruction–generation tension explicit by aligning the learned VAE code to a frozen vision foundation model while retaining pixel, perceptual, adversarial, and KL objectives [Yao et al., 2025]. This is still a learned compressor: the VFM is a target, not the inference-time encoder. A related staged family starts from VFM features and progressively adapts an adapter and decoder. ALIGNTOK's three stages—latent alignment, perceptual alignment, and decoder refinement—separate semantic preservation from fine-detail recovery rather than asking one joint loss to solve both at once [Chen et al., 2025].

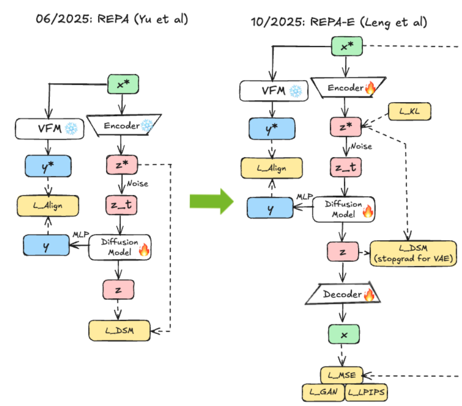
The optimization conflict is easy to state. Suppose a nuisance variable $n$ affects pixels but not category. A representation objective may reward

$$I(z;y) \text{ large}, \qquad I(z;n) \text{ small},$$

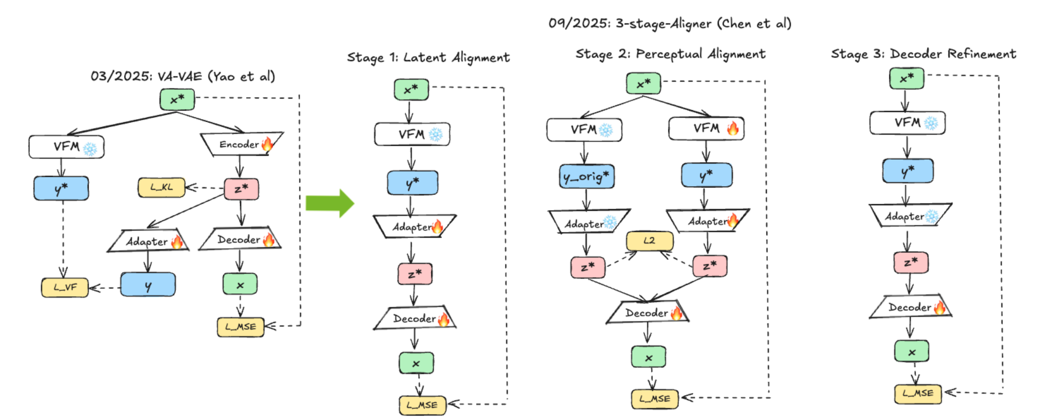
whereas an invertible decoder needs enough information about $n$ to reconstruct $x$. A practical tokenizer can allocate low-frequency or semantic information to an aligned subspace and residual detail to another, but this is an architectural compromise rather than a proof that semantics and fidelity are naturally identical.




(a) VA-VAE moves VFM alignment from denoiser features into tokenizer training.




(b) REPA-E jointly schedules representation, autoencoding, and denoising objectives while routing gradients.




(c) A staged alternative: latent alignment, perceptual adaptation, then decoder refinement.

Figure 6: Ways to shape the tokenizer with representations: VA-VAE [Yao et al., 2025], REPA-E [Leng et al., 2025], and the three-stage AlignTok recipe [Chen et al., 2025]. These systems can look similar at inference while optimizing different gradient graphs. Author-created Excalidraw diagrams.

### 5.2 Representation autoencoders

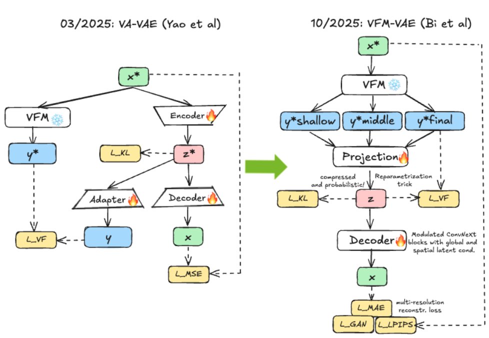
A representation autoencoder (RAE) uses a pretrained encoder $E_{\mathrm{VFM}}$ as the image code and trains a decoder $D_\psi$ to reconstruct pixels. The prior learns

$$z_0 = E_{\mathrm{VFM}}(x), \qquad \mathcal{L}_{\mathrm{prior}} = \mathbb{E}\|v_\theta(z_t, t) - u_t\|^2.$$

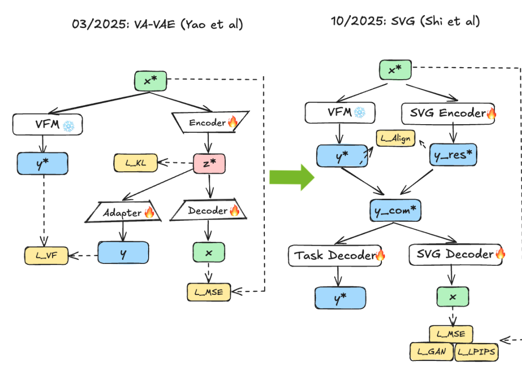
The promise is compelling: replace a low-level VAE code with a semantically organized representation and inherit the scaling of foundation-model pretraining. The cost is that the code may be high-dimensional, anisotropic, lossy in local detail, or concentrated near a curved manifold. A decoder trained only on exact encoder outputs may also be brittle to the approximate samples produced by the prior.

The original RAE work couples a frozen pretrained encoder with a learned pixel decoder and adapts the generative architecture to the resulting wide latent [Zheng et al., 2025]. Scaling studies show that conclusions depend on DiT width and encoder dimension [Tong et al., 2026]. RAEv2 aggregates several encoder layers, compares 27 encoders, and combines the latent with internal representation guidance; its main lesson is that RAE and REPA are complementary, not mutually exclusive [Singh et al., 2026]. LV-RAE restores low-level information and explicitly trains decoder robustness around the feature manifold [Liu et al., 2026].

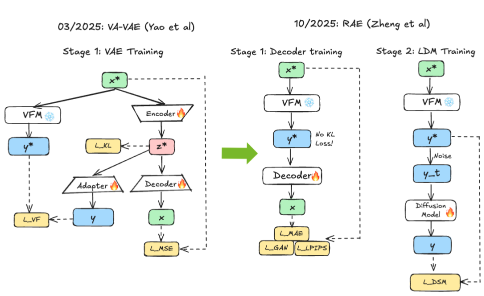
Two intermediate designs explain why "replace the VAE" is not a single operation. VFM-VAE merges several frozen VFM layers, compresses them through a regularized latent, and reconstructs progressively at multiple scales [Bi et al., 2025]. SVG instead combines a frozen semantic feature with a learned residual branch for details that the VFM discarded [Shi et al., 2025]. RAE removes both the additional learned semantic alignment and the mandatory residual encoder: a frozen pretrained representation is the code, and the burden shifts to the decoder and prior.



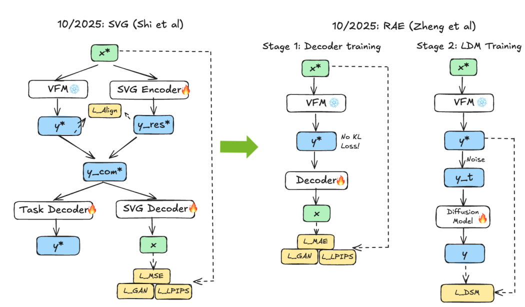

(a) VFM-VAE aggregates several frozen VFM layers into a compact regularized code.




(b) SVG preserves a frozen semantic branch and learns a residual detail branch.




(c) RAE uses the pretrained representation directly and learns the inverse decoder.




(d) From the hybrid SVG interface to the simpler but much wider RAE interface.

Figure 7: The Phase-3 design progression from aligned learned codes to pretrained-feature latents [Yao et al., 2025, Bi et al., 2025, Shi et al., 2025, Zheng et al., 2025]. These are intervention-point comparisons, not claims that every later method strictly subsumes every earlier one. Author-created Excalidraw diagrams.

FAE is a useful boundary case. It keeps the VFM frozen but compresses its patch features using one self-attention layer plus a projection. A six-layer feature decoder reconstructs the original VFM feature field using feature MSE and a small KL term; a separate pixel decoder is trained on noisy original features and then refined on reconstructed features [Gao et al., 2026b]. Only after this staged adaptation is a generative prior learned. It is therefore neither one projection and done nor a fully joint generator. Its claim is that a very small trainable bottleneck can preserve useful VFM geometry while restoring a conventional, compact generative interface.

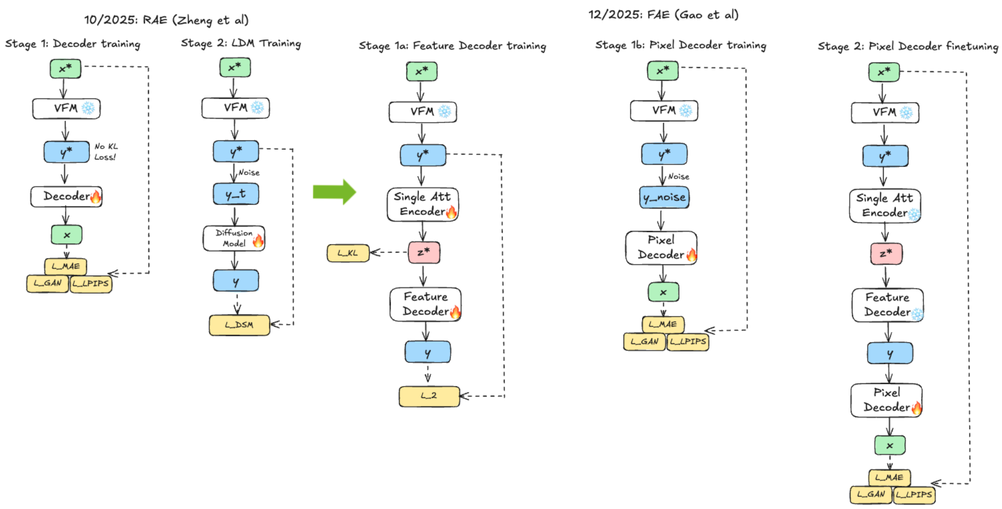



Figure 8: FAE compresses frozen VFM features through a shallow trainable adapter, reconstructs the original feature field with a deeper feature decoder, and separately trains a pixel decoder before fitting the latent prior [Gao et al., 2026b]. The diagram makes clear why "one layer" describes the feature adapter, not the whole system. Author-created Excalidraw diagram.

**Discrete and spherical variants.** IDEAL aligns discrete codes with shallow and deep VFM features [Chen et al., 2026]; DRAE uses hyperspherical codes to reduce sensitivity to feature magnitude [Ma et al., 2026]. RAE-AR addresses sequential generation by normalizing codes and injecting Gaussian noise during autoregressive training [Yu et al., 2026]. Here exposure bias has its sequence-model meaning: training sees ground-truth previous codes, whereas generation conditions on its own imperfect samples. This differs from poor reconstruction on clean features and from geometric path mismatch, even though all three can produce off-manifold decoder inputs.

# 6 Jointly Learning the Latent Interface

## 6.1 Unified Latents: rate as a design variable

UNIFIED LATENTS uses a deterministic encoder followed by fixed Gaussian noise, a diffusion prior whose minimum noise matches the encoder noise, and a diffusion decoder [Heek et al., 2026]. The prior is trained with an unweighted diffusion ELBO, while reweighting the decoder ELBO controls which information remains in the latent. This construction makes rate explicit rather than treating latent dimension as its proxy.

For any auxiliary code distribution $q_\phi(z \mid x)$ and prior $p_\theta(z)$,

$$\mathbb{E}_{p(x)} D_{\mathrm{KL}}(q_\phi(z \mid x) \| p_\theta(z)) = \underbrace{I_q(X;Z)}_{\text{information rate}} + \underbrace{D_{\mathrm{KL}}(q_\phi(z) \| p_\theta(z))}_{\text{aggregate mismatch}}, \tag{7}$$

where $q_\phi(z) = \int p(x) q_\phi(z \mid x)\, dx$. The expected prior cost upper-bounds mutual information because the second term is non-negative. It simultaneously charges for information in each code and for a prior that fails to match the aggregate posterior. UL operationalizes this relation with diffusion-based likelihood terms.

| Method | Representation source | Prior gradient to clean encoder | Anti-collapse / control | Final prior protocol | Most precise description |
|---|---|---|---|---|---|
| REPA-E | VAE encoder aligned to frozen VFM | ✗ | alignment and reconstruction | jointly scheduled DiT | encoder adaptation with routed gradients |
| UL | learned noisy continuous code | ✓ | fixed noise and variational rate | strongest result retrains prior | joint interface learning with explicit rate |
| UNITE | shared Generative Encoder/registers | ✗ (best setting) | stop-gradient and mode sharing | shared latent denoiser | shared parameters, selectively routed gradient |
| GenFirst | learned continuous code | ✓ | entropy term and curriculum | fresh prior also evaluated | prior-shaped representation with collapse control |

Table 2: Why end-to-end is not a sufficient method description.

The nuance matters: encoder, prior, and decoder are co-trained in the first stage, but UL's strongest reported generation configuration freezes the learned autoencoder and trains a fresh prior afterward. This is evidence that joint training can learn a useful interface, not evidence that a single training job is always optimal.

### 6.2 UNITE: shared weights with routed gradients

UNITE uses one Generative Encoder in two modes: tokenization of clean data and denoising of corrupted latent tokens [Duggal et al., 2026]. Register tokens create a compact code, and the same core weights participate in representation formation and latent denoising. Yet the best configuration stops the denoising gradient at the clean latent. That stop-gradient does not make the tasks independent: the denoising mode still updates shared parameters, which subsequently changes tokenization. The coupling is *through shared weights across modes*, not through direct differentiation of the prior loss through the clean-code computation.

This distinction explains why two statements can both be true: UNITE is an end-to-end shared model, and naive direct latent gradients can be harmful. The architecture couples the tasks while the stop-gradient controls a high-variance or collapse-prone route. UNITE also reports experiments on images, QM9 molecules, and MP20 crystals, making it unusually relevant to cross-domain transfer.

### 6.3 GenFirst: give the prior a gradient, then prevent collapse

GenFirst explicitly allows the generative prior to shape the encoder [Zheng et al., 2026]. Its diagnosis is that naive joint likelihood training contains pressure to reduce the entropy of the aggregate code because a concentrated distribution is easier for the prior. A schematic objective is

$$\mathcal{J} = \underbrace{\lambda_{\text{rec}}\mathcal{L}_{\text{rec}}}_{\text{retain information}} + \underbrace{\lambda_{\text{fit}}\mathcal{L}_{\text{prior}}}_{\text{fit aggregate codes}} - \underbrace{\lambda_H H(q_\phi(z))}_{\text{oppose collapse}}, \tag{8}$$

combined with a generation-first schedule: prioritize a modelable latent before tightening reconstruction. The entropy term is not decorative. Without a countervailing force, the prior objective can be improved by mapping diverse $x$ to similar $z$.

GenFirst and UL therefore share the ambition to make rate and generation pressure part of representation learning, but their mechanisms differ. UL derives a variational rate-control construction with fixed encoder noise; GenFirst foregrounds entropy balance and curriculum. GenFirst, too, evaluates a freshly trained prior after learning the autoencoder, so end-to-end latent discovery and final prior training should be reported separately.

## 7 What Makes a Representation Generative?

We propose reporting six partly independent properties.

**1. Reconstruction fidelity.** Measure distortion in pixels and perceptual features, but also editing consistency and fine-structure recovery. Clean reconstruction tests $D(E(x))$, not $D(\tilde{z})$ for prior samples $\tilde{z}$.

**2. Semantic organization.** Linear probes, retrieval, segmentation, and robustness reveal whether downstream factors are accessible. They do not establish invertibility. Moreover, spatial and global semantics play different roles: patchwise denoiser alignment rewards spatial correspondence, while a latent prior may benefit from globally meaningful organization.

**3. Information rate.** Latent dimension alone is not rate. A continuous $64 \times 64 \times 1024$ tensor can carry little information if heavily noised, or enormous information at high precision. Report an entropy or coding proxy, encoder noise, quantization, and the prior cross-entropy where possible. Equation (7) provides a principled decomposition.

**4. Distributional modelability.** Whitening, channel scale, token outliers, covariance condition number, and tail behavior affect optimization. A generator can fail on a perfectly reconstructive latent simply because the aggregate distribution is ill-conditioned. Conversely, a low-dimensional smooth code may be easy to model while discarding details the decoder cannot recover.

**5. Geometry and spectrum.** If features concentrate on a sphere or another manifold, Euclidean interpolation can traverse low-density interiors. Hyperspherical methods such as SPHERE ENCODER and DRAE make the geometry explicit [Yue et al., 2026, Ma et al., 2026]. RJF sharpens this into a path-design claim for pretrained representation latents: projecting Gaussian noise onto the feature sphere repairs the endpoints, but a linear interpolant still follows an off-manifold chord; SLERP instead follows a geodesic and its velocity remains tangent to the sphere [Kumar and Patel, 2026]. In the paper's controlled protocol, endpoint projection alone improved FID only marginally, whereas the full geometry-aware method improved much more. This is evidence that geometry can explain failures otherwise attributed to insufficient Transformer width; it is not proof that capacity never matters, because architecture, objective, and solver are changed together in the full comparison. Appendix A.6 gives the derivation.

Frequency organization is another view. SPECTRUM MATCHING proposes that a diffusion-friendly encoder should expose a suitably flattened power spectrum and that decoder training should pair image- and latent-frequency perturbations [Ning et al., 2026].

The exact foundation is Parseval's identity. For an orthonormal transform $F$,

$$\begin{aligned} \|x - \hat{x}\|_2^2 &= \|F(x - \hat{x})\|_2^2 \\ &= \sum_k \underbrace{|(Fx)_k - (F\hat{x})_k|^2}_{\text{error at frequency } k}. \end{aligned} \tag{9}$$

Thus pixel MSE decomposes exactly by frequency. It does *not* follow as a theorem that a particular flattened spectrum is sufficient or optimal for diffusion. The stronger spectrum-matching hypothesis is an empirically supported design hypothesis. It is complementary to rate and manifold geometry, not a replacement for them.

**6. Decoder robustness.** At inference, the decoder sees samples from the learned prior, not exact encoder outputs. Let $\delta = \tilde{z} - E(x)$. Locally,

$$D(E(x) + \delta) \approx D(E(x)) + J_D(E(x))\delta.$$

Even small latent error becomes large pixel error when the decoder Jacobian has high gain along likely prior-error directions. Noise injection, drifted-latent training, consistency losses, and explicit Jacobian control all target this mismatch. The right perturbation distribution should resemble actual prior errors rather than isotropic noise by default.

| Property | Useful measurements | Common proxy failure | Representative interventions |
|---|---|---|---|
| Fidelity | PSNR/SSIM, rFID, LPIPS, edit recovery | clean recon ignores sampled latents | residual detail paths, richer decoder |
| Semantics | probes, retrieval, dense tasks | global probe hides spatial mismatch | REPA, multi-layer VFM features |
| Rate | nats/bits, prior NLL, encoder noise | tensor size is not information | UL weighting/noise, quantization |
| Modelability | prior loss, covariance, tails, convergence | rFID can improve while gFID worsens | normalization, compact bottleneck |
| Geometry / spectrum | norm/radius, geodesic tests, PSD | Euclidean metrics ignore support | spherical codes, spectrum matching |
| Robustness | drifted-latent recon, Jacobian gain | $D(E(x))$ is in-distribution only | noisy-feature or sampled-code training |

Table 3: A minimum diagnostic panel for generative representations.

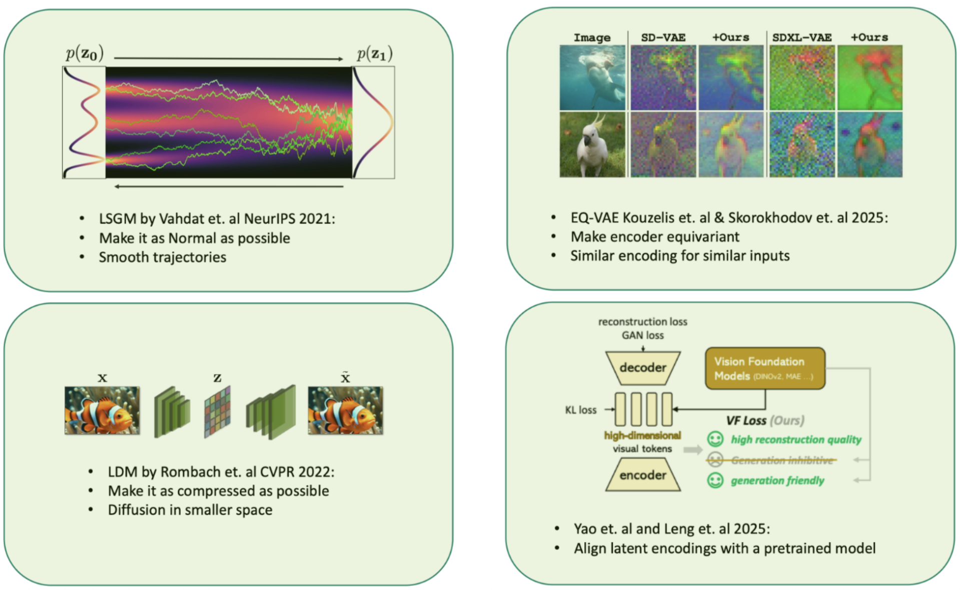



Figure 9: Earlier intuitions about a "good latent space" expressed as four design motifs: smooth prior geometry, equivariant or perturbation-stable codes, computational compression, and semantic alignment. The six-property framework in Table 3 extends this picture by separating rate, modelability, geometry, and decoder robustness. Author-created Excalidraw composition; small panels adapted from the cited source works remain copyright of their original authors and publishers and are excluded from this paper's CC BY license.

## 8 Native and Self-Supervised Generative Representations

### 8.1 Sphere Encoder and iterative refinement

SPHERE ENCODER learns an encoder and decoder whose aggregate code is conditionally uniform on a hypersphere, so a normalized Gaussian sample is already a valid code [Yue et al., 2026]. It can generate in one decoder pass. Its optional few-step procedure then re-encodes the decoded image, adds method-specific tangent-space noise during spherification, and decodes again. This is not an RAE: the code is learned for generation rather than inherited from a pretrained semantic encoder. It is also not a diffusion prior, and the refinement loop is not a deep equilibrium model: the method executes a chosen finite number of encoder–decoder passes and does not define or implicitly differentiate a solved fixed point. The closest description is recurrent learned projection toward the image manifold.

### 8.2 Latent Forcing: order the information reveal

LATENT FORCING keeps pixel-space generation but jointly diffuses pixels and pretrained semantic features with separate time schedules [Baade et al., 2026]. Denote noisy pixels $x_{t_x}$ and noisy semantic features $r_{t_r}$. A joint network predicts both flows:

$$\mathcal{L} = \mathbb{E}\big[\|v_x(x_{t_x}, r_{t_r}, t_x, t_r) - u_x\|^2 + \lambda_r\|v_r(x_{t_x}, r_{t_r}, t_x, t_r) - u_r\|^2\big] .$$

By choosing $t_r$ to resolve semantics before $t_x$ resolves pixels, the trajectory exposes a coarse semantic plan before high-frequency synthesis. This is trajectory design, not tokenizer learning. The paper's strongest schedule should not be generalized into a universal rule without considering modality and architecture.

### 8.3 Self-Flow: the teacher is generated internally

SELF-FLOW removes the external semantic teacher by creating asymmetric views of the same noisy sample [Chefer et al., 2026]. Tokens receive heterogeneous noise times. An exponential-moving-average (EMA) teacher observes a cleaner version, while the student observes mixed noise and predicts teacher features in addition to the flow target. Schematically,

$$h^- = f_{\bar{\theta}}(x_{\min(t,s)}, \min(t, s)), \tag{10}$$

$$\mathcal{L}_{\text{SF}} = \underbrace{\|v_\theta(x_t, t) - u_t\|^2}_{\text{flow}} + \lambda \underbrace{\left[1 - \cos(g(h_\theta(x_t, t, s)), \text{sg}(h^-))\right]}_{\text{internal feature target}} . \tag{11}$$

The EMA, stop-gradient, and cleaner-context asymmetry prevent the target from trivially chasing the student. The method spans image, video, audio, multimodal, and action experiments, with modality-specific schedules. It removes dependence on an external VFM; it does not imply the absence of an autoencoder or tokenizer in every implementation.

**Product-scale evidence.** Black Forest Labs states that FLUX.3 builds on Self-Flow and uses a shared flow-matching backbone across image, video, audio, and action generation [Black Forest Labs, 2026]. This is relevant evidence that the design family is being scaled, but the public report describes evaluations as preliminary and does not disclose enough detail for independent reproduction. It should be cited as a first-party system announcement, not counted as a controlled research comparison.

## 9 Beyond Images

### 9.1 Language and unified multimodal models

LDLM jointly adapts pretrained language features, a latent diffusion model, and a decoder [Meshchaninov et al., 2026]. It uses a hidden-state reconstruction loss, a warm-up that delays difficult coupling, adaptive timestep sampling, and noise on decoder inputs. These stabilizers mirror the visual literature: control the gradient path, teach the decoder about inference-time drift, and allocate optimization effort where the prior is weakest.

REPR-ALIGN takes the opposite direction, adapting an autoregressive language model toward diffusion through representation alignment rather than fully retraining it [Peng et al., 2026a]. UNIAR argues that a shared context-visual tokenizer is central to unified autoregressive understanding and generation [Peng et al., 2026b]. Controlled 2026 studies report that generation and understanding can transfer when they require shared knowledge, but also identify asymmetric interference when both are forced through identical computation paths [Zhu et al., 2026, Wu et al., 2026]. Unified should therefore be decomposed into shared vocabulary, shared latent, shared trunk, shared objective, and shared inference procedure.

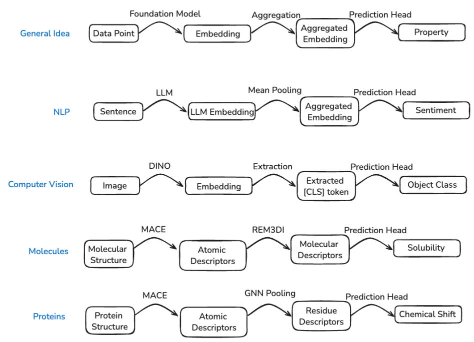
### 9.2 Molecules, proteins, and 3D scenes

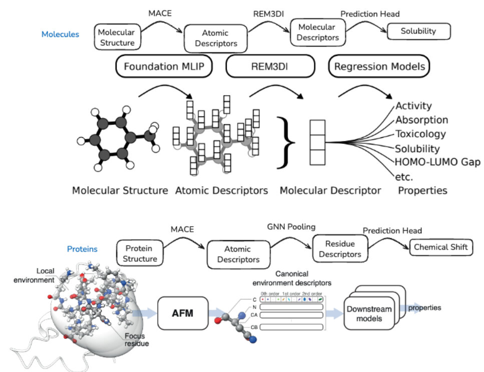
Structural domains sharpen the interface problem because validity is geometric. Rotations and translations should not change identity; atom order may be arbitrary; chirality matters; structures vary in size; and a small local coordinate error can propagate globally. A semantic image feature can discard texture harmlessly, whereas a protein code that discards stereochemistry is chemically wrong.

UNITE reports the same shared tokenization/denoising mechanism on QM9 molecules and MP20 crystals as well as images [Duggal et al., 2026]. KANZI uses a flow-matching decoder as a compact protein-structure tokenizer and reports strong reconstruction with a much smaller model than specialized geometric tokenizers [Dilip et al., 2026]. ONEWORLD builds a 3D unified representation autoencoder from pretrained 3D features, adds cross-view consistency, and trains on both clean and drifted representations [Gao et al., 2026a]. These examples support the transfer of *principles*—rate, geometry, and decoder robustness—not the literal reuse of image architectures.

Atomistic foundation models provide a second bridge. MACE embeddings transfer from energy/force pretraining to molecular and protein property prediction [Batatia et al., 2025, Wedig et al., 2025, Bojan et al., 2025]. As in language and vision, a local feature field must be aggregated into a task-level representation. The generative use cases are more tentative: representation alignment has been explored for force prediction and conformation generation [Pinede et al., 2025], while direct conditioning on atomistic embeddings has not uniformly improved structure prediction [Corley et al., 2025]. This distinction between transferable features, aligned generators, and architectural inspiration prevents a downstream probe result from being overread as a generative result.

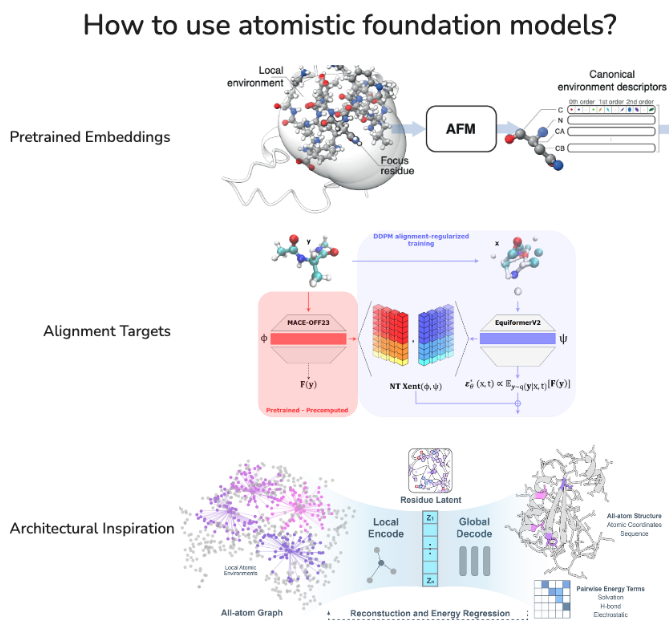
Protein modeling also clarifies the difference between representation recurrence and generative self-conditioning. AlphaFold recycling repeatedly feeds predicted structure-derived features back through the network to refine one prediction [Jumper et al., 2021]; ESMFold uses a related folding trunk around language-model features [Lin et al., 2023]. RFdiffusion uses diffusion self-conditioning in structure generation [Watson et al., 2023]. The generic diffusion technique predates RFdiffusion: Analog Bits explicitly introduced self-conditioning of a model's previous clean-data estimate for discrete diffusion [Chen et al., 2023]. Recycling, self-conditioning, and iterative decoding all feed predictions forward, but differ in state, objective, and stochastic semantics. Appendix B makes the distinction precise.




(a) Local MACE descriptors are aggregated into molecular representations for property prediction.

(b) The same local-to-global pattern across NLP, vision, molecules, and proteins.




(c) Three roles for atomistic foundation models: pretrained embeddings, alignment targets, and architectural inspiration.

Figure 10: How the visual representation–generation taxonomy transfers to atomistic systems [Batatia et al., 2025, Wedig et al., 2025, Bojan et al., 2025, Pinede et al., 2025]. Author-created Excalidraw compositions. Embedded molecular and protein renderings or small source-paper panels remain copyright of their original authors and publishers and are excluded from this paper's CC BY license.

## 10 Evaluation and Reporting

### 10.1 Why headline FID is insufficient

Generation FID depends on training data, resolution, sample count, guidance scale, number of function evaluations, sampler, model size, augmentation, and evaluation implementation. Reconstruction FID depends on the decoder and on whether stochasticity is used. Reporting a lower number across papers does not identify which representation is better unless these variables are controlled.

At minimum, a representation paper should report:

1. clean reconstruction and reconstruction under realistic latent drift;
2. unconditional or class/text-conditional generation with and without guidance;
3. latent rate or a defensible proxy, spatial size, channel count, normalization, and encoder noise;
4. prior convergence curves at matched compute, not only the best checkpoint;
5. semantic and dense/spatial probes where relevant;
6. all stop-gradients, frozen modules, EMA targets, warm-ups, and stage boundaries; and
7. final-prior protocol, especially whether a new prior is trained after joint latent learning.

### 10.2 Ablations that answer causal questions

An informative ablation changes one interface property while holding capacity and compute fixed. Examples include whitening versus raw features; identical decoder with clean versus drifted-latent training; equal-rate latents with different dimensions; direct prior-to-encoder gradient versus stop-gradient under the same shared trunk; and aligned targets with preserved versus shuffled spatial structure. Scaling only the denoiser may hide a difficult latent rather than repair it.

The most useful negative results are boundary conditions. If representation alignment stops helping after early training, that locates its role as curriculum. If a VFM latent helps a wide DiT but not a narrow one, architecture and representation are coupled. If fresh-prior retraining improves a jointly learned system, that distinguishes interface discovery from final density estimation. Such findings are more reusable than a single champion configuration.

## 11 Synthesis: What the 2026 Wave Changes

The 2026 papers do not replace the earlier story; they split several overloaded concepts.

**End-to-end is a family of gradient graphs.** REPA-E aligns the encoder but blocks the diffusion gradient. UNITE shares weights across tokenization and denoising but performs best with a stopped clean code. UL co-trains encoder, prior, and decoder under a variational rate construction, then obtains its strongest generation with a fresh prior. GenFirst deliberately transmits prior pressure to the encoder and introduces entropy and curriculum to prevent collapse. These are not cosmetic variants.

**Representation quality is multidimensional.** SPECTRUM MATCHING adds frequency organization; spherical methods add explicit topology; UL adds rate; LV-RAE and OneWorld add decoder robustness; RAE-AR adds sequential exposure bias. No one diagnostic subsumes the others.

**Internal teachers reduce dependence on external encoders.** Self-Flow shows that heterogeneous-noise prediction and an EMA teacher can create useful targets within the generator. Latent Forcing shows that a pretrained semantic track can instead order the generation trajectory. Their shared lesson is controlled information asymmetry, not that all representations should be discarded.

**Unification benefits from specialization with communication.** Shared representations can transfer knowledge between understanding and generation, but fully identical paths can interfere. A common interface with mode-specific heads, schedules, or routed gradients may be more effective than one undifferentiated state.

## 12 Limitations and Open Problems

This survey reflects a rapidly moving literature and a September 2026 cutoff. Many works have not completed peer review; code availability and reported status may change. We emphasize primary papers and official project material, but cannot normalize every training and evaluation protocol. Our taxonomy assigns a dominant mechanism, which necessarily compresses hybrids.

Several research questions appear durable:

1. Can latent rate, prior difficulty, and decoder robustness be measured in a common evaluation rather than inferred from architecture?
2. Which prior-error distribution should a decoder train against, and can it be estimated online without destabilizing joint learning?
3. When does a curved manifold require a geometry-aware probability path rather than normalization or increased model width?
4. Can a joint objective adapt semantic features while certifiably preserving downstream capabilities?
5. What is the correct information ordering for video, audio, language, and 3D structures?
6. How should unified systems allocate shared versus specialized capacity as modalities and tasks scale?

## 13 Conclusion

The field is converging on a simple but demanding idea: a generative latent is an interface, not a by-product. Alignment methods change internal optimization; tokenizer methods decide which information is retained; RAEs inherit semantic geometry and its defects; joint methods let the prior reshape the code but must control collapse; native self-supervision creates targets from the generative process itself.

The practical rule is to inspect three objects together: the information carried by $z$, the errors made by the prior, and the decoder's response to those errors. Then trace every gradient. This view explains why reconstruction, semantics, spectrum, geometry, and rate can each matter without any one becoming the universal answer. It also scales beyond images: the constraints change, but the interface-design problem remains.

**Author's note.** This article represents the author's personal scholarly synthesis and does not necessarily reflect the views of NVIDIA. 

## References

Alan Baade, Eric Ryan Chan, Kyle Sargent, Changan Chen, Justin Johnson, Ehsan Adeli, and Li Fei-Fei. Latent forcing: Reordering the diffusion trajectory for pixel-space image generation. *arXiv preprint arXiv:2602.11401*, 2026. URL `https://arxiv.org/abs/2602.11401`.

Shaojie Bai, J. Zico Kolter, and Vladlen Koltun. Deep equilibrium models. In *Advances in Neural Information Processing Systems*, 2019. URL `https://arxiv.org/abs/1909.01377`.

Ilyes Batatia et al. A foundation model for atomistic materials chemistry. *The Journal of Chemical Physics*, 163(18), 2025. doi: 10.1063/5.0292065.

Tianyang Bi, Xiaoyu Zhang, Yujie Lu, and Nanning Zheng. Vision foundation models can be good tokenizers for latent diffusion models. *arXiv preprint arXiv:2510.18457*, 2025. URL `https://arxiv.org/abs/2510.18457`.

Black Forest Labs. FLUX 3: Real world models – towards multimodal flow models as the backbone of visual intelligence. Company blog, July 2026. URL `https://bfl.ai/blog/flux-3`. Accessed 2026-09-07.

Milan Bojan, Sai Vedula, Anirudh Maddipatla, et al. Representing local protein environments with atomistic foundation models. *arXiv preprint arXiv:2505.23354*, 2025. URL https://arxiv.org/abs/2505.23354.

Mathilde Caron, Hugo Touvron, Ishan Misra, Herve Jegou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *IEEE/CVF International Conference on Computer Vision*, 2021. URL https://arxiv.org/abs/2104.14294.

Hila Chefer, Patrick Esser, Dominik Lorenz, Dustin Podell, Vikash Raja, Vinh Tong, Antonio Torralba, and Robin Rombach. Self-supervised flow matching for scalable multi-modal synthesis. In *International Conference on Machine Learning*, 2026. URL https://arxiv.org/abs/2603.06507.

Bowei Chen, Sai Bi, Hao Tan, et al. AlignTok: Aligning visual foundation encoders to tokenizers for diffusion models. *arXiv preprint arXiv:2509.25162*, 2025. URL https://arxiv.org/abs/2509.25162.

Ting Chen, Ruixiang Zhang, and Geoffrey Hinton. Analog bits: Generating discrete data using diffusion models with self-conditioning. *International Conference on Learning Representations*, 2023. URL https://arxiv.org/abs/2208.04202.

Yitong Chen, Zijie Diao, Junke Wang, et al. IDEAL: In-depth alignment makes a discrete representation autoencoder. *arXiv preprint arXiv:2606.11096*, 2026. URL https://arxiv.org/abs/2606.11096.

Nathan Corley, Simon Mathis, Rohan Krishna, et al. Accelerating biomolecular modeling with AtomWorks and RF3. *bioRxiv*, 2025. doi: 10.1101/2025.08.14.670328.

Ranjita Dilip, Eric Zhang, Anushka Varshney, and David Van Valen. Flow autoencoders are effective protein tokenizers. In *International Conference on Learning Representations*, 2026. URL https://openreview.net/forum?id=5p9uled7JM.

Shivam Duggal, Xingjian Bai, Zongze Wu, Richard Zhang, Eli Shechtman, Antonio Torralba, Phillip Isola, and William T. Freeman. End-to-end training for unified tokenization and latent denoising. *arXiv preprint arXiv:2603.22283*, 2026. URL https://arxiv.org/abs/2603.22283.

Sensen Gao, Zhaoqing Wang, Qihang Cao, Dongdong Yu, Changhu Wang, Tongliang Liu, Mingming Gong, and Jiawang Bian. OneWorld: Taming scene generation with 3d unified representation autoencoder. *arXiv preprint arXiv:2603.16099*, 2026a. URL https://arxiv.org/abs/2603.16099.

Yuan Gao, Chen Chen, Tianrong Chen, and Jiatao Gu. One layer is enough: Adapting pretrained visual encoders for image generation. *arXiv preprint arXiv:2512.07829*, 2026b. URL https://arxiv.org/abs/2512.07829. Version 2 manuscript dated August 2026.

Kaiming He, Xinlei Chen, Saining Xie, Yanghao Li, Piotr Dollar, and Ross Girshick. Masked autoencoders are scalable vision learners. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022. URL https://arxiv.org/abs/2111.06377.

Jonathan Heek, Emiel Hoogeboom, Thomas Mensink, and Tim Salimans. Unified latents (UL): How to train your latents. *arXiv preprint arXiv:2602.17270*, 2026. URL https://arxiv.org/abs/2602.17270.

Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, 2020. URL https://arxiv.org/abs/2006.11239.

Sepp Hochreiter and Jurgen Schmidhuber. Long short-term memory. *Neural Computation*, 9(8): 1735–1780, 1997.

John Jumper, Richard Evans, Alexander Pritzel, et al. Highly accurate protein structure prediction with AlphaFold. *Nature*, 596:583–589, 2021. doi: 10.1038/s41586-021-03819-2.

Diederik P. Kingma and Max Welling. Auto-encoding variational bayes. In *International Conference on Learning Representations*, 2014. URL https://arxiv.org/abs/1312.6114.

Amandeep Kumar and Vishal M. Patel. Learning on the manifold: Unlocking standard diffusion transformers with representation encoders. *arXiv preprint arXiv:2602.10099*, 2026. URL https://arxiv.org/abs/2602.10099.

Xingjian Leng, Jaskirat Singh, Yunzhong Hou, Zhenchang Xing, Saining Xie, and Liang Zheng. REPA-E: Unlocking VAE for end-to-end tuning with latent diffusion transformers. *arXiv preprint arXiv:2504.10483*, 2025. URL https://arxiv.org/abs/2504.10483.

Zeming Lin, Halil Akin, Roshan Rao, et al. Evolutionary-scale prediction of atomic-level protein structure with a language model. *Science*, 379(6637):1123–1130, 2023. doi: 10.1126/science.ade2574.

Yaron Lipman, Ricky T. Q. Chen, Heli Ben-Hamu, Maximilian Nickel, and Matt Le. Flow matching for generative modeling. In *International Conference on Learning Representations*, 2023. URL https://arxiv.org/abs/2210.02747.

Siyu Liu, Chujie Qin, Hubery Yin, et al. Improving reconstruction of representation autoencoder. *arXiv preprint arXiv:2602.08620*, 2026. URL https://arxiv.org/abs/2602.08620.

Tianren Ma, Lin Long, Chuyan Chen, Mu Zhang, Junbo Zhao, Tong Zhang, and Qixiang Ye. dRAE: Representation autoencoder with hyper-spherical codes. *arXiv preprint arXiv:2607.22148*, 2026. URL https://arxiv.org/abs/2607.22148.

Viacheslav Meshchaninov, Alexander Shabalin, Egor Chimbulatov, Nikita Gushchin, Ilya Koziev, Alexander Korotin, and Dmitry Vetrov. How to train your latent diffusion language model jointly with the latent space. *arXiv preprint arXiv:2605.07933*, 2026. URL https://arxiv.org/abs/2605.07933.

Mang Ning, Mingxiao Li, Le Zhang, Lanmiao Liu, Matthew B. Blaschko, Albert Ali Salah, and Itir Onal Ertugrul. Spectrum matching: A unified perspective for superior diffusability in latent diffusion. *arXiv preprint arXiv:2603.14645*, 2026. URL https://arxiv.org/abs/2603.14645.

Maxime Oquab, Timothee Darcet, Theo Moutakanni, et al. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024. URL https://arxiv.org/abs/2304.07193.

William Peebles and Saining Xie. Scalable diffusion models with transformers. In *IEEE/CVF International Conference on Computer Vision*, 2023. URL https://arxiv.org/abs/2212.09748.

Fred Zhangzhi Peng, Alexis Fox, Anru R. Zhang, and Alexander Tong. Don't retrain, align: Adapting autoregressive LMs to diffusion LMs via representation alignment. *arXiv preprint arXiv:2605.06885*, 2026a. URL https://arxiv.org/abs/2605.06885.

Wujian Peng, Lingchen Meng, Yuxuan Cai, et al. Unified multimodal autoregressive modeling with shared context-visual tokenizer is key to unification. *arXiv preprint arXiv:2606.18249*, 2026b. URL https://arxiv.org/abs/2606.18249.

Lucas Pinede, Sijia Yang, Juno Nam, and Rafael Gómez-Bombarelli. Unifying force prediction and molecular conformation generation through representation alignment. In *ICML Workshop on Generative AI and Biology*, 2025. URL https://openreview.net/forum?id=yzkHGHvC74.

Alec Radford, Jong Wook Kim, Chris Hallacy, et al. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, 2021. URL https://arxiv.org/abs/2103.00020.

Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Bjorn Ommer. High-resolution image synthesis with latent diffusion models. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022. URL https://arxiv.org/abs/2112.10752.

Minghao Shi et al. Latent diffusion model without variational autoencoder. *arXiv preprint arXiv:2510.15301*, 2025. URL https://arxiv.org/abs/2510.15301.

Jaskirat Singh, Xingjian Leng, Zongze Wu, Liang Zheng, Richard Zhang, Eli Shechtman, and Saining Xie. What matters for representation alignment: Global information or spatial structure? *arXiv preprint arXiv:2512.10794*, 2025. URL https://arxiv.org/abs/2512.10794.

Jaskirat Singh, Boyang Zheng, Zongze Wu, Richard Zhang, Eli Shechtman, and Saining Xie. Improved baselines with representation autoencoders. *arXiv preprint arXiv:2605.18324*, 2026. URL https://arxiv.org/abs/2605.18324.

Ivan Skorokhodov, Sharath Girish, Benran Hu, Willi Menapace, Yanyu Li, Rameen Abdal, Sergey Tulyakov, and Aliaksandr Siarohin. Improving the diffusability of autoencoders. *arXiv preprint arXiv:2502.14831*, 2025. URL https://arxiv.org/abs/2502.14831.

Yang Song, Jascha Sohl-Dickstein, Diederik P. Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *International Conference on Learning Representations*, 2021. URL https://arxiv.org/abs/2011.13456.

Shengbang Tong, Boyang Zheng, Ziteng Wang, et al. Scaling text-to-image diffusion transformers with representation autoencoders. In *International Conference on Learning Representations*, 2026. URL https://arxiv.org/abs/2601.16208.

Arash Vahdat, Karsten Kreis, and Jan Kautz. Score-based generative modeling in latent space. In *Advances in Neural Information Processing Systems*, 2021. URL https://arxiv.org/abs/2106.05931.

Aaron van den Oord, Oriol Vinyals, and Koray Kavukcuoglu. Neural discrete representation learning. In *Advances in Neural Information Processing Systems*, 2017. URL https://arxiv.org/abs/1711.00937.

Ziqiao Wang, Wangbo Zhao, Yuhao Zhou, et al. REPA works until it doesn't: Early-stopped, holistic alignment supercharges diffusion training. *arXiv preprint arXiv:2505.16792*, 2025. URL https://arxiv.org/abs/2505.16792.

Joseph L. Watson, David Juergens, Nathaniel R. Bennett, et al. De novo design of protein structure and function with RFdiffusion. *Nature*, 620:1089–1100, 2023. doi: 10.1038/s41586-023-06415-8.

Sammy Wedig, Rokas Elijošius, Christoph Schran, and Lars L. Schaaf. REM3DI: Learning smooth, chiral 3d molecular representations from equivariant atomistic foundation models. In *NeurIPS Workshop on Symmetry and Geometry in Neural Representations*, 2025. URL https://openreview.net/forum?id=jOmZsvXoK5.

Chen Wei, Karttikeya Mangalam, Po-Yao Huang, Yanghao Li, Haoqi Fan, Hu Xu, Huiyu Wang, Cihang Xie, Alan Yuille, and Christoph Feichtenhofer. Diffusion models as masked autoencoders. In *IEEE/CVF International Conference on Computer Vision*, pages 16284–16294, 2023. URL https://openaccess.thecvf.com/content/ICCV2023/html/Wei_Diffusion_Models_as_Masked_Autoencoders_ICCV_2023_paper.html.

Ge Wu, Shen Zhang, Ruijing Shi, et al. Representation entanglement for generation: Training diffusion transformers is much easier than you think. *arXiv preprint arXiv:2507.01467*, 2025. URL https://arxiv.org/abs/2507.01467.

Penghao Wu, Haiwen Diao, Weichen Fan, Lewei Lu, Dahua Lin, and Ziwei Liu. Uncovering understanding-generation synergy in native unified multimodal models: From representation, task to system. *arXiv preprint arXiv:2609.01607*, 2026. URL https://arxiv.org/abs/2609.01607.

Weilai Xiang, Hongyu Yang, Di Huang, and Yunhong Wang. Denoising diffusion autoencoders are unified self-supervised learners. In *IEEE/CVF International Conference on Computer Vision*, pages 15802–15812, 2023. URL https://openaccess.thecvf.com/content/ICCV2023/html/Xiang_Denoising_Diffusion_Autoencoders_are_Unified_Self-supervised_Learners_ICCV_2023_paper.html.

Xingyi Yang and Xinchao Wang. Diffusion model as representation learner. In *IEEE/CVF International Conference on Computer Vision*, pages 18938–18949, 2023. URL `https://openaccess.thecvf.com/content/ICCV2023/html/Yang_Diffusion_Model_as_Representation_Learner_ICCV_2023_paper.html`.

Jinwei Yao, Bin Yang, and Xinchao Wang. Reconstruction vs. generation: Taming optimization dilemma in latent diffusion models. *arXiv preprint arXiv:2501.01423*, 2025. URL `https://arxiv.org/abs/2501.01423`.

Hu Yu, Hang Xu, Jie Huang, Zeyue Xue, Haoyang Huang, Nan Duan, and Feng Zhao. RAE-AR: Taming autoregressive models with representation autoencoders. *arXiv preprint arXiv:2604.01545*, 2026. URL `https://arxiv.org/abs/2604.01545`.

Sihyun Yu, Sangkyung Kwak, Huiwon Jang, Jongheon Jeong, Jonathan Huang, Jinwoo Shin, and Saining Xie. Representation alignment for generation: Training diffusion transformers is easier than you think. *arXiv preprint arXiv:2410.06940*, 2024. URL `https://arxiv.org/abs/2410.06940`.

Kaiyu Yue, Menglin Jia, Ji Hou, and Tom Goldstein. Image generation with a sphere encoder. *arXiv preprint arXiv:2602.15030*, 2026. URL `https://arxiv.org/abs/2602.15030`.

Shilong Zhang, He Zhang, Zhifei Zhang, et al. Both semantics and reconstruction matter: Making representation encoders ready for text-to-image generation and editing. *arXiv preprint arXiv:2512.17909*, 2025. URL `https://arxiv.org/abs/2512.17909`.

Boyang Zheng, Nanye Ma, Shengbang Tong, and Saining Xie. Diffusion transformers with representation autoencoders. *arXiv preprint arXiv:2510.11690*, 2025. URL `https://arxiv.org/abs/2510.11690`.

Guangting Zheng, Yiyuan Zhang, Tao Yang, Yunpeng Chen, Rui Zhu, Jiajun Deng, and Yanyong Zhang. GenFirst: Generation before reconstruction for stable end-to-end latent generative modeling. *arXiv preprint arXiv:2608.29335*, 2026. URL `https://arxiv.org/abs/2608.29335`.

Yubo Zhu, Zhehan Kan, Jingyi Yang, et al. When does visual generation help visual understanding in unified multimodal models? *arXiv preprint arXiv:2608.22174*, 2026. URL `https://arxiv.org/abs/2608.22174`.

# A Detailed Derivations

## A.1 The VAE bound, line by line

Starting from the marginal likelihood and multiplying by one,

$$\begin{aligned}
\log p(x) &= \log \int p(x,z)\,dz && \text{(marginalize } z\text{)} && (12)\\
&= \log \int q_\phi(z \mid x)\frac{p(x,z)}{q_\phi(z \mid x)}\,dz && \text{(importance identity)} && (13)\\
&= \log \mathbb{E}_{q_\phi(z|x)}\left[\frac{p_\psi(x \mid z)p(z)}{q_\phi(z \mid x)}\right] && && (14)\\
&\geq \mathbb{E}_{q_\phi(z|x)}\left[\log p_\psi(x \mid z) + \log p(z) - \log q_\phi(z \mid x)\right] && \text{(Jensen)} && (15)\\
&= \underbrace{\mathbb{E}_q[\log p_\psi(x \mid z)]}_{\text{decoder likelihood}} - \underbrace{D_{\mathrm{KL}}(q_\phi(z \mid x)\|p(z))}_{\text{rate and regularity}}. && && (16)
\end{aligned}$$

The gap equals $D_{\mathrm{KL}}(q_\phi(z \mid x)\|p(z \mid x))$, so the bound is tight when the approximate posterior matches the true posterior. Intuitively, the encoder sends a message $z$; reconstruction rewards useful bits, while the KL charges for messages that are surprising under the prior.

### A.2 Why the expected prior cost contains mutual information

Let $q(x, z) = p_{\text{data}}(x)q_\phi(z \mid x)$. Add and subtract $\log q_\phi(z)$:

$$\begin{aligned}
\mathbb{E}_{p(x)} D_{\text{KL}}(q(z \mid x) \| p(z)) &= \mathbb{E}_{q(x,z)} \left[ \log \frac{q(z \mid x)}{p(z)} \right] &(17)\\
&= \mathbb{E}_{q(x,z)} \left[ \underbrace{\log \frac{q(z \mid x)}{q(z)}}_{\text{mutual information}} + \underbrace{\log \frac{q(z)}{p(z)}}_{\text{aggregate mismatch}} \right] &(18)\\
&= I_q(X; Z) + D_{\text{KL}}(q(z) \| p(z)). &(19)
\end{aligned}$$

The second term vanishes only when the prior exactly matches the aggregate posterior. A stronger learned prior can reduce aggregate mismatch without reducing the information rate. This is why latent dimension and prior loss should not be used interchangeably.

### A.3 Noise prediction, clean prediction, and velocity

From $z_t = \alpha_t z_0 + \sigma_t \epsilon$, any two of $z_t, z_0, \epsilon$ determine the third:

$$\hat{z}_0 = \frac{z_t - \sigma_t \hat{\epsilon}}{\alpha_t}, \qquad \hat{\epsilon} = \frac{z_t - \alpha_t \hat{z}_0}{\sigma_t}.$$

Define velocity $v_t = \alpha_t \epsilon - \sigma_t z_0$. Then

$$\begin{bmatrix} z_t \\ v_t \end{bmatrix} = \begin{bmatrix} \alpha_t & \sigma_t \\ -\sigma_t & \alpha_t \end{bmatrix} \begin{bmatrix} z_0 \\ \epsilon \end{bmatrix}, \qquad \begin{bmatrix} z_0 \\ \epsilon \end{bmatrix} = \begin{bmatrix} \alpha_t & -\sigma_t \\ \sigma_t & \alpha_t \end{bmatrix} \begin{bmatrix} z_t \\ v_t \end{bmatrix}.$$

The matrix is orthogonal when $\alpha_t^2 + \sigma_t^2 = 1$. Thus the parameterizations encode equivalent targets at population optimum, but their finite-capacity weighting and numerical conditioning differ by timestep.

### A.4 Why naive prior pressure can collapse a learned code

Consider a deterministic encoder $z = E_\phi(x)$ and a flexible prior trained by negative log likelihood,

$$\mathcal{L}_{\text{prior}} = -\mathbb{E}_x \log p_\theta(E_\phi(x)).$$

If the decoder term is weak, mapping every input to a constant $z_\star$ makes the aggregate code maximally concentrated and easy for the prior to fit. A reconstruction loss resists because one code cannot reconstruct all inputs. An entropy reward resists directly:

$$\mathcal{J} = \lambda_{\text{rec}} \mathcal{L}_{\text{rec}} + \lambda_{\text{fit}} \mathcal{L}_{\text{prior}} - \lambda_H H(q_\phi(z)).$$

This toy argument does not prove a particular algorithm stable; it explains why letting the prior shape the encoder requires an opposing information-preservation mechanism.

### A.5 Spectrum matching and the limit of Parseval

Let $F$ be an orthonormal discrete Fourier or cosine transform, so $F^\top F = I$. Then

$$\begin{aligned}
\|x - \hat{x}\|_2^2 &= (x - \hat{x})^\top (x - \hat{x}) &(20)\\
&= (x - \hat{x})^\top F^\top F (x - \hat{x}) \qquad \text{(orthonormality)} &(21)\\
&= \|F(x - \hat{x})\|_2^2 = \sum_k |(Fx)_k - (F\hat{x})_k|^2. &(22)
\end{aligned}$$

If a binary frequency mask $M$ is applied consistently, the loss becomes

$$\|M \odot F(x - \hat{x})\|_2^2 = \sum_{k: M_k = 1} |(Fx)_k - (F\hat{x})_k|^2.$$

This exactly justifies frequency-resolved reconstruction. It does not identify the optimal latent power spectral density. That step requires assumptions about the data, model class, noise schedule, and optimization, or empirical validation.

### A.6 Why Euclidean chords leave a hypersphere

The second reading-group deck asks whether a representation generator fails because the denoiser is too narrow or because its probability path has the wrong geometry. The two hypotheses are not mutually exclusive. The following calculation isolates the geometric part used by RJF [Kumar and Patel, 2026].

Let $x, \epsilon \in \mathbb{S}^{d-1}$ be unit vectors and define their angle by

$$x^\top \epsilon = \cos\Omega, \qquad 0 < \Omega < \pi.$$

Standard conditional flow matching uses the Euclidean chord $z_t = (1-t)x + t\epsilon$. Its squared radius is

$$\begin{aligned}
\|z_t\|_2^2 &= \big((1-t)x + t\epsilon\big)^\top \big((1-t)x + t\epsilon\big) && \text{(definition of the norm)} && (23)\\
&= (1-t)^2\|x\|^2 + t^2\|\epsilon\|^2 + 2t(1-t)x^\top\epsilon && \text{(bilinearity)} && (24)\\
&= (1-t)^2 + t^2 + 2t(1-t)\cos\Omega && \text{(unit endpoints and angle).} && (25)
\end{aligned}$$

At the midpoint,

$$\begin{aligned}
\|z_{1/2}\|_2^2 &= \frac{1}{2}(1+\cos\Omega) && \text{(substitute } t = 1/2\text{)} && (26)\\
&\approx \frac{1}{2}, \qquad \|z_{1/2}\|_2 \approx \frac{1}{\sqrt{2}} && \text{(near-orthogonality in high dimension).} && (27)
\end{aligned}$$

The approximation uses concentration: independently oriented high-dimensional unit vectors typically have $x^\top\epsilon \approx 0$, hence $\Omega \approx \pi/2$. For a sphere of radius $R = \sqrt{d}$, the same calculation gives a midpoint radius near $\sqrt{d/2}$, even though both endpoints have radius $\sqrt{d}$. Normalizing only the noise endpoint therefore does not repair the intervening path.

Spherical linear interpolation replaces the chord by

$$z_t^{\text{slerp}} = a_t x + b_t \epsilon, \quad a_t = \frac{\sin((1-t)\Omega)}{\sin\Omega}, \quad b_t = \frac{\sin(t\Omega)}{\sin\Omega}. \tag{28}$$

Set $A = (1-t)\Omega$ and $B = t\Omega$, so $A + B = \Omega$. Then

$$\begin{aligned}
\|z_t^{\text{slerp}}\|^2 &= a_t^2 + b_t^2 + 2a_t b_t \cos\Omega && \text{(same bilinear expansion)} && (29)\\
&= \frac{\sin^2 A + \sin^2 B + 2\sin A \sin B \cos(A+B)}{\sin^2(A+B)} && \text{(substitute } a_t, b_t, \Omega = A+B\text{)} && (30)\\
&= 1 && \text{(trigonometric identity).} && (31)
\end{aligned}$$

Thus every intermediate point remains on the unit sphere. On a 2D unit circle with $x = (1,0)$, $\epsilon = (0,1)$, and $\Omega = \pi/2$, the Euclidean midpoint is $(1/2, 1/2)$ with radius $1/\sqrt{2}$, whereas the SLERP midpoint is $(1/\sqrt{2}, 1/\sqrt{2})$ with radius one. The distinction is chord versus arc, not merely Gaussian versus normalized noise.

Differentiating Equation (28) gives the target tangent velocity

$$u_t^{\mathcal{M}} = \dot{z}_t^{\text{slerp}} = \frac{\Omega}{\sin\Omega}\left[\cos(t\Omega)\epsilon - \cos((1-t)\Omega)x\right] \qquad \text{(chain rule).} \tag{32}$$

Because the path has constant norm,

$$\begin{aligned}
0 &= \frac{d}{dt}\|z_t^{\text{slerp}}\|^2 = 2(z_t^{\text{slerp}})^\top \dot{z}_t^{\text{slerp}} && \text{(differentiate the invariant)} && (33)\\
\implies \quad & (z_t^{\text{slerp}})^\top u_t^{\mathcal{M}} = 0 && \text{(tangent-space condition).} && (34)
\end{aligned}$$

A network prediction $v_\theta$ can be made tangent by removing its radial component,

$$v_\theta^{\tan} = v_\theta - (v_\theta^\top z_t) z_t.$$

At sampling time, an Euler step would leave the sphere. The exponential map instead rotates along a great circle:

$$z_{t+\Delta t} = \cos(\|v\|\Delta t) z_t + \sin(\|v\|\Delta t)\frac{v}{\|v\|}, \qquad v \perp z_t.$$

RJF additionally asks how an instantaneous transverse velocity error $w$ affects a later endpoint. For a sphere of radius $R$, sectional curvature is $K = 1/R^2$. A Jacobi field with $J(0) = 0$ and $D_\tau J(0) = w$ has endpoint magnitude, for remaining arc length $L = R(1-t)\Omega$,

$$\|J(1)\| = \|w\| \frac{\sin(\sqrt{K}L)}{\sqrt{K}} \qquad \text{(Jacobi equation on constant curvature)} \tag{35}$$

$$= \|w\| L \frac{\sin((1-t)\Omega)}{(1-t)\Omega} \qquad \text{(substitute } K = 1/R^2,\ L = R(1-t)\Omega). \tag{36}$$

Relative to flat-space propagation $\|w\|L$, the differential of the exponential map scales transverse error by $\operatorname{sinc}((1-t)\Omega)$. Squaring the endpoint error yields the method's weight

$$\lambda(t,\Omega) = \operatorname{sinc}^2((1-t)\Omega), \qquad \mathcal{L}_{\mathrm{RJF}} = \mathbb{E}\big[\lambda(t,\Omega)\|v_\theta^{\tan} - u_t^{\mathcal{M}}\|^2\big]\,. \tag{37}$$

This weighting is specific to the paper's sphere, endpoint, and time convention; it is not a universal consequence of using SLERP. The exact facts are the chord-radius calculation, norm preservation, and tangency. The claim that these changes remove the practical need for width scaling is empirical and should be tested under matched architectures and compute.

### A.7 Decoder robustness as a directional problem

For prior sample error $\delta$, first-order propagation gives

$$\mathbb{E}\|D(z+\delta) - D(z)\|_2^2 \approx \mathbb{E}\left[\delta^\top J_D(z)^\top J_D(z)\delta\right]\,.$$

If $\Sigma_\delta = \mathbb{E}[\delta\delta^\top]$, then

$$\mathbb{E}\|J_D\delta\|^2 = \operatorname{tr}\big(J_D^\top J_D \Sigma_\delta\big)\,.$$

Isotropic noise training uses $\Sigma_\delta = \sigma^2 I$ and penalizes average Jacobian energy. Actual prior errors can be anisotropic, concentrating along difficult directions. Sampling the current prior or fitting a local error covariance can therefore train the decoder more efficiently than indiscriminate isotropic perturbation.

## B Self-Conditioning, Recycling, Looping, and Recurrence

These labels describe a shared computational motif—feed an intermediate prediction back into later computation—but not a shared probabilistic object.

**Self-conditioning in diffusion.** At step $t$, a denoiser consumes a previous estimate of the clean sample,

$$\hat{x}_0^{(t)} = f_\theta(x_t, t, \operatorname{sg}(\hat{x}_0^{(t+1)})).$$

Analog Bits used this technique for discrete-data diffusion before RFdiffusion adopted it for protein structures [Chen et al., 2023, Watson et al., 2023]. The stop-gradient means the previous estimate acts as an auxiliary condition during ordinary training. It changes the parameterized reverse kernel while preserving a valid generative procedure if the augmented state and training/inference conditioning are defined consistently. What is usually lost is the simplicity of a Markov kernel written only as $p_\theta(x_{t-1} \mid x_t)$: the implementation depends on cached model state. It does not automatically invalidate diffusion, but likelihood or ELBO statements derived for the unaugmented Markov chain cannot be quoted without modification.

**Recycling.** AlphaFold recycling maps a current structure and pair/single representations back into the folding trunk to refine the same prediction [Jumper et al., 2021]. It is supervised iterative inference, not a stochastic diffusion-time transition. Parameter sharing makes the unfolded computation recurrent. Training may unroll a limited number of recycles and stop gradients through earlier iterations for memory or stability.

| Mechanism | Feedback state | Index being iterated | Typical training gradient | Probabilistic role |
|---|---|---|---|---|
| Diffusion self-conditioning | previous $\hat{x}_0$ | noise/sampling time | often stopped through cached estimate | augments reverse-kernel context |
| Protein recycling | structure, single/pair features | refinement recycle | finite unroll or stopped earlier recycle | iterative supervised inference |
| Looped Transformer | hidden state or tokens | depth step | BPTT or truncation | architecture dependent |
| DEQ | equilibrium hidden state | root-solver iteration | implicit differentiation | implicit layer, not inherently generative |
| Sphere Encoder refinement | re-encoded decoded sample | refinement loop | method-specific finite loop | moves sample toward learned data manifold |

Table 4: Related computation patterns with different states, objectives, and theoretical commitments.

**Looping and generic recurrence.** A looped Transformer reapplies a block $h_{k+1} = F_\theta(h_k, x)$, often with shared weights. Backpropagation through time (BPTT) differentiates through the unrolled chain:

$$\frac{\partial \mathcal{L}}{\partial \theta} = \sum_{k=0}^{K-1} \frac{\partial \mathcal{L}}{\partial h_K} \left( \prod_{j=k+1}^{K-1} \frac{\partial h_{j+1}}{\partial h_j} \right) \frac{\partial h_{k+1}}{\partial \theta}.$$

The Jacobian product explains exploding or vanishing gradients and motivated gated recurrent networks [Hochreiter and Schmidhuber, 1997]. Truncated BPTT stops some paths and optimizes a surrogate.

**Deep equilibrium models.** A deep equilibrium model defines $h^\star = F_\theta(h^\star, x)$ and solves for the fixed point rather than storing all iterations [Bai et al., 2019]. Implicit differentiation gives

$$\frac{dh^\star}{d\theta} = \left( I - \frac{\partial F_\theta}{\partial h}\bigg|_{h^\star} \right)^{-1} \frac{\partial F_\theta}{\partial \theta}\bigg|_{h^\star}.$$

DEQs are therefore a mathematically important ancestor of learned iterative refinement, but most recycling or self-conditioning systems are finite unrollings and do not enforce convergence to a fixed point. The useful connection is shared-weight computation and stability analysis; equality of names is not evidence of an equilibrium model.

## C Method Cards for the 2026 Additions

**Unified Latents [Heek et al., 2026].** Intervention: learned continuous code. Coupling: joint encoder/prior/decoder first stage. Distinctive control: fixed encoder noise matched to the prior's minimum noise and a reweighted decoder ELBO. Caveat: strongest generation includes fresh-prior retraining.

**UNITE [Duggal et al., 2026].** Intervention: one register-token Generative Encoder serves tokenizer and latent-denoising modes. Coupling: shared parameters, with stopped clean-code gradient in the best setting. Evidence: images plus QM9 and MP20. Caveat: shared weights should not be described as unrestricted gradient sharing.

**GenFirst [Zheng et al., 2026].** Intervention: jointly learned latent shaped directly by the prior. Coupling: active prior-to-encoder gradient. Distinctive control: entropy counterterm and generation-before-reconstruction schedule. Caveat: code was pending at the survey cutoff, and a fresh final prior remains part of the evaluation.

**FAE [Gao et al., 2026b].** Intervention: compact bottleneck on frozen VFM features. Coupling: staged. Distinctive control: one-layer encoder, deeper feature decoder, and separate pixel decoder exposed to noisy features. Caveat: one layer describes the adapter, not the entire autoencoding system.

**Spectrum Matching [Ning et al., 2026].** Intervention: encoder latent spectrum and decoder frequency training. Coupling: tokenizer-level. Distinctive control: encoding spectrum matching, decoding spectrum matching, and a Difference-of-Gaussians alignment target. Caveat: the algebraic decomposition is exact; the proposed preferred spectrum is empirical.

**RJF [Kumar and Patel, 2026].** Intervention: geometry-aware flow over normalized pretrained features. Coupling: frozen representation encoder plus a learned Riemannian latent prior. Distinctive control: spherical endpoint noise, SLERP paths, tangent velocity prediction, exponential-map integration, and Jacobi weighting. Caveat: the chord calculation is exact, but the claim that geometry rather than width is the dominant bottleneck rests on the paper's experimental protocol.

**Sphere Encoder [Yue et al., 2026].** Intervention: learned hyperspherical code with a simple uniform prior. Coupling: learned generative autoencoder. Distinctive control: conditional uniformity, direct decoding, and optional finite encoder–decoder refinement with tangent-space noise. Caveat: neither a pretrained-feature RAE, a diffusion prior, nor an equilibrium model.

**Latent Forcing [Baade et al., 2026].** Intervention: an auxiliary semantic trajectory in pixel diffusion. Coupling: joint dual-track denoising with separately scheduled times. Distinctive control: semantics resolve before pixels. Caveat: no learned tokenizer is implied.

**Self-Flow and FLUX.3 [Chefer et al., 2026, Black Forest Labs, 2026].** Intervention: internal EMA representation target from a cleaner view. Coupling: native self-supervised flow training. Distinctive control: heterogeneous token timesteps and cross-time prediction. Caveat: the academic method has technical detail; FLUX.3 is a first-party deployment report with preliminary evaluation.

## D Expanded Molecular and Structural Case Study

The six-property framework changes meaning in structural domains:

| | |
|---|---|
| **Fidelity** | Bond lengths, local frames, clashes, chirality, and long-range topology matter; a perceptually plausible structure may still be chemically invalid. |
| **Semantics** | Functional family, fold, secondary structure, or energy landscape can be more relevant than image-style class labels. |
| **Rate** | Variable-size structures complicate per-sample comparisons. Report bits or nats per atom/residue as well as token count. |
| **Modelability** | Equivariant coordinate distributions and discrete sequence/structure mixtures require different priors. |
| **Geometry** | SE(3) symmetry, periodic boundary conditions, and manifold-valued rotations are first-class constraints rather than optional regularizers. |
| **Robustness** | Small local-frame or torsion errors can accumulate along a chain; decoder training should reflect the error modes of the structure prior. |

This makes the analogy to visual RAEs useful but incomplete. A pretrained 3D encoder can provide semantics, yet its invariances may remove precisely the degrees of freedom required for generation. Cross-view or equivariant consistency can preserve identity while avoiding arbitrary coordinate frames. Drifted-latent decoder training is especially important when validity depends on coordinated global constraints.

UNITE's cross-domain experiments suggest that shared register-token interfaces can survive a change in data type [Duggal et al., 2026]. Kanzi suggests that flow-based decoding itself can be an efficient tokenizer design [Dilip et al., 2026]. OneWorld demonstrates pretrained 3D features, cross-view consistency, and exposure to drifted representations [Gao et al., 2026a]. The open question is not whether RAE for molecules works in the abstract, but which invariances, rates, and perturbations correspond to the downstream structural generator.

## E Reading-Group Guide

A 60-minute discussion can be organized around five questions:

1. **10 min: What is the interface?** Derive Equations (1) and (7); separate distortion, rate, and aggregate prior mismatch.
2. **12 min: Where does representation enter?** Use Figure 4 to contrast denoiser alignment, tokenizer shaping, RAEs, and native systems.
3. **15 min: What does end-to-end mean?** Trace Figure 2 and Table 2; ask which gradient each stabilizer blocks.
4. **13 min: What makes a latent modelable?** Derive chord versus SLERP using Appendix A.6, then debate whether rate, geometry, spectrum, decoder robustness, or model capacity explains each result.
5. **10 min: Does the analogy transfer?** Compare image latents with language and molecular structures, then identify broken invariances.

**Suggested core reading.** Read REPA [Yu et al., 2024] for auxiliary alignment, RAE [Zheng et al., 2025] for pretrained-feature latents, and either UL [Heek et al., 2026] or GenFirst [Zheng et al., 2026] for joint latent learning. Add RJF [Kumar and Patel, 2026] for the geometry-versus-capacity debate, Spectrum Matching [Ning et al., 2026] for diagnostics, and Self-Flow [Chefer et al., 2026] for native self-supervision. DDAE [Xiang et al., 2023] is a useful historical counterpoint: it asks what representation is already inside a diffusion model. UNITE [Duggal et al., 2026] is the best bridge to non-image domains in this set.